

# Ajitkumar Senthil Kumar

Applied AI Engineer | Agentic Systems & RAG | Python • TypeScript

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Applied AI Engineer with **3 years** of software engineering experience, specializing in **Agentic AI and RAG systems**. Builds production AI applications using LangGraph, FastAPI, and Next.js. Researching Deep learning and Reinforcement learning methods for adaptive AI systems.

## EDUCATION

University at Buffalo, SUNY | *M.S., Data Science*

2024 – 2026

Anna University | *B.E., Mechanical Engineering*

2017 – 2021

## TECHNICAL SKILLS

**Languages:** Python, SQL, TypeScript, JavaScript

**AI / ML Core:** PyTorch, PyTorch Lightning, Scikit-learn, Deep Learning, Computer Vision, LLM Fine-Tuning

**LLM & Agents:** LangGraph, LangChain, LlamaIndex, Agentic RAG, MCP, Mastra, A2A Protocol, Neo4j, pgvector

**AI-Augmented Development:** Cursor, Claude Code, OpenAI Codex, Spec-Driven Development

**Backend & Cloud:** FastAPI, OpenClaw, PostgreSQL, Supabase, Redis, Snowflake, AWS, Docker, NVIDIA Triton

**Frontend & Tools:** React, Next.js, Astro.js, Tailwind CSS, GitHub Actions, CI/CD

## EXPERIENCE

Research Assistant, AI Systems — A2IL Lab, University at Buffalo —

Mar 2026 – Present

MyVictor — University Chatbot —

- Delivered a **production conversational AI platform for university advising**, backed by **~300 automated tests**, integrating asynchronous APIs, stateless SSE streaming, and non-blocking conversation persistence while preserving the existing public API contract.
- Reduced **LLM calls to zero for common list/count queries** by executing structured database queries directly; strengthened answer grounding with **Neo4j + Qdrant hybrid retrieval**, reranking, and self-correcting query repair evaluated across **37 QA cases**.
- Enforced **approved-user access and private conversation history** through allowlisted authentication and ownership-scoped data access, with an **audited, read-only admin view** for platform oversight.
- Deployed **MyVictor on OpenShift**, standardizing application delivery through integrated container orchestration, routing, and release management within the university's centrally managed infrastructure.

Faculty Monitoring System —

- Built a **research intelligence platform** tracking **80+ UB faculty**, integrating **12 academic and federal sources** through automated ingestion, entity resolution, and provenance-aware **PostgreSQL/Supabase** storage.
- Delivered **FastAPI + Next.js** workflows combining **retrieval-grounded LLM synthesis**, human-reviewable weekly reports on publications and grants, and automated **Gmail/Slack alerts** through **Composio**.

Software Developer — Human CloudSoft Pvt Ltd —

May 2023 – Jun 2024

- Developed **18 RESTful API endpoints**, enabling centralized inventory tracking, stock updates, and record retrieval across **4 operational teams**.
- Improved **inventory accuracy by 35%** by replacing manual recordkeeping with **React.js interfaces** and **ASP.NET MVC** business logic for validated stock transactions.
- Reduced **inventory audit time by 40%** by developing **T-SQL stored procedures**, optimizing indexed queries, and streamlining database-driven stock reconciliation.
- Strengthened **API security and data integrity** through server-side validation, input sanitization, and Postman testing, while maintaining **SQL Server persistence** and supporting **AWS deployment**.

Junior Software Developer — Klicknet Info Services Pvt Ltd —

Apr 2022 – Apr 2023

- Developed and delivered **reusable React.js and JavaScript Functional Components** for client-facing web applications, standardizing responsive layouts by **almost 70%** and reducing duplicate UI implementation.
- Integrated **Flask REST APIs** with frontend workflows, implementing **unit and API-to-database integration tests** to validate business logic, data persistence, and error handling before release.

## PROJECTS

Nexomnis AI — Multi-LLM Council, Agentic RAG & Agent Orchestration 🌀 —

April 2026

- Architected a **FastAPI + LangGraph + Next.js** agentic platform with Supabase/pgvector, retrieval tracing, structured observability, and **cost-aware orchestration across 5 LLM providers**.
- Built **NexAsk's 4-role adversarial council** with atomic-claim verification and adaptive **pgvector/BM25 retrieval**, HyDE, query decomposition, reranking, and prosecutor-triggered re-retrieval.
- Implemented **NexTask's LangGraph execution pipeline** with Composio, Tavily, and Mem0, persisting council traces as preference data for downstream **DPO/GRPO alignment experiments and evaluation**.

MIDAS — Multi-Agent Medical Vision & Evidence-Grounded AI 🌀 —

November 2025

- Developed an **8-class skin-lesion classification system** using **EfficientNet-B0** and **Grad-CAM**, confidence-scored predictions and visual explanations to support interpretation of model outputs.
- Achieved **~90% Recall@10** and **75–82% Precision@3** in internal retrieval evaluations by combining **hybrid search and reranking** to retrieve relevant dermatology evidence with Multi-Agentic RAG Architecture for citation validation and post-generation claim verification.
- Unified **3 role-based workspaces—patient, clinician, and admin** with server-side access controls, connecting patient image submissions and reports, clinician knowledge ingestion, and administrative user management and system monitoring.